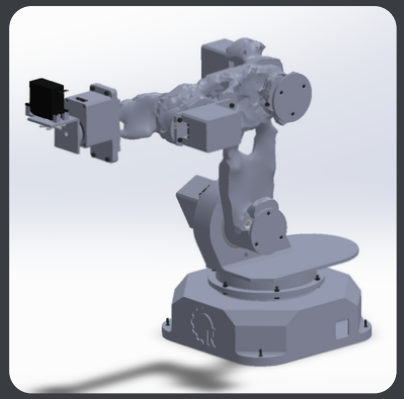
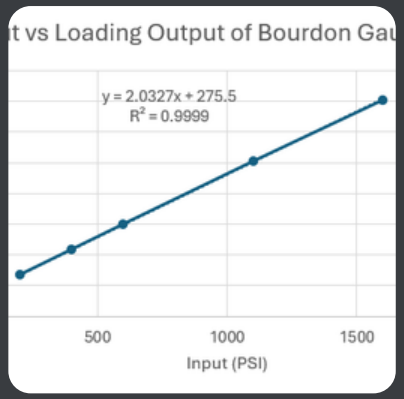
Nathaniel Cavazos

Mechanical Engineering Portfolio

Mechanical Engineer Student at California State University, Long Beach
B.S. Mechanical Engineering | Expected May 2028





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Portfolio - nathaniel-cavazos.com

3 – Technical Skills & Organizations

4 – Six Degrees of Freedom Robotic Arm

4.1 - Requirements & System Architecture
4.2 - Design
4.3 - System Analysis
4.4 - Iterations
4.5 - Testing
4.6 - Software Development
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5 – Bourdon Gauge & Load Cell Calibration

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5.2 - Sample Calculations
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5.5 - Conclusion

CAD/CAE:

- SolidWorks
  - Static Analysis
  - Topology Analysis
- Ansys
  - Static Analysis
  - Topology Analysis
- Siemens NX
- MATLAB
- Simulink

Engineering Software:

- Common Components Workbench
- RSLinx
- BOOTP

PCB Design:

- KiCad

Embedded System:

- STM32
- SPI
- UART
- I2C
- Sensor Integration

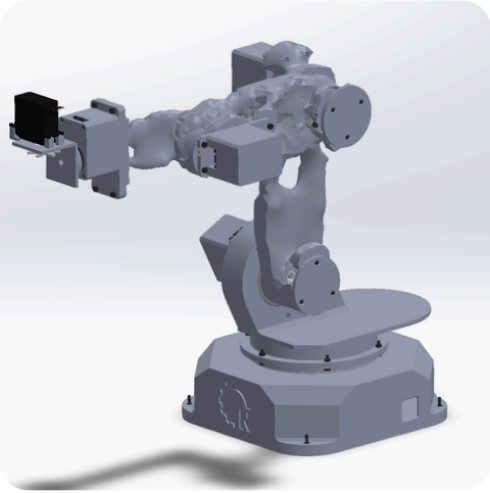
Relevant Coursework:

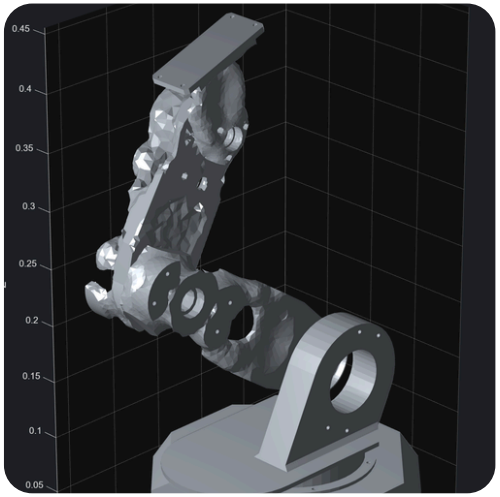
- Applied Math
- Electricity & Magnetism
- Statics
- Dynamics
- Deformable Bodies
- Fluid Dynamics
- Engineering Materials
- Numerical Methods

Organizations:

- Beach Launch Team
- SASE
- Pi Tau Sigma

6-Degree-Of-Freedom (DOF) Robotic Arm





Project Overview: For Version 1, I focused on designing, optimizing, and simulating a 6-DOF robotic arm. I developed this project to explore mechanical design, structural analysis, kinematics, and automation. Using SOLIDWORKS, I designed the mechanical components and joint assemblies, and I used ANSYS to perform structural analysis and topology optimization. I then used MATLAB's Robotics System Toolbox to develop a kinematic model and simulate the first four joints' movement. Future versions will focus on physically building the robotic arm and developing a functional end effector.

Tools: SolidWorks • Ansys • MATLAB • Robotics System Toolbox

Requirements:

- 6 Independant controller axes
- Inverse & Forward Kinematics System
- Joint Angle Limiters
- Homing System
- Safety Fault Systems
- Max Load of 4lb for 1 minute & 30 seconds
- Minimum Reach of 16 in
- Mechanical Grippers
- A Factor of Safety >= 2.0
- Optimized Strength-to-Weight Ratio
- End Deflector Position < 5mm
- Yield Strength on All Joints< 8 MPa

System Architecture:

For version 1, I divided the project into three main stages: mechanical design, structural optimization, and kinematic simulation. I first used SOLIDWORKS to develop the robotic arm's components and joint assemblies. I then used ANSYS to analyze the structural performance of the links and optimize their geometry. After completing the design, I imported the optimized links into MATLAB's Robotics System Toolbox to define the joint positions, rotation axes, and transformations. This allowed me to simulate the robotic arm's movement and evaluate different joint configurations.

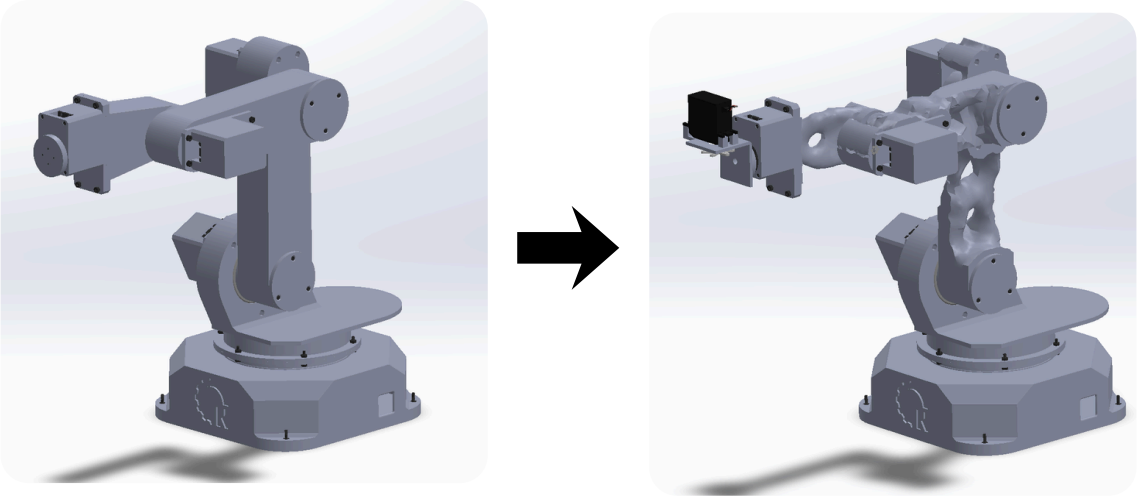
Future versions will focus on integrating the physical actuators, implementing joint controls, and developing the mechanical gripper.

SolidWorks - Mechanical Design ➡ **ANSYS - Structural Analysis & Optimization** ➡ **MATLAB - Kinematic Modeling & Simulation**

Nathaniel Cavazos
Mechanical Engineer at California State University Long Beach

Design Consideration:

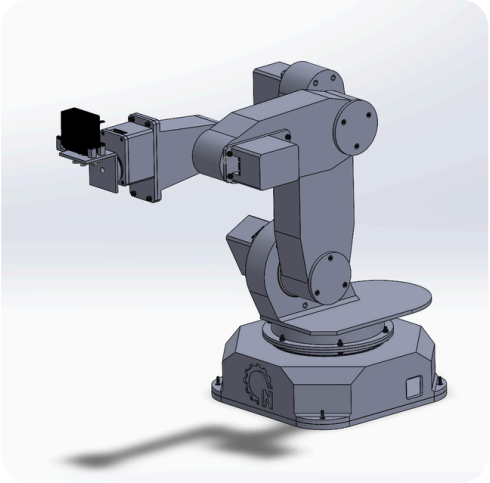
Design 1:



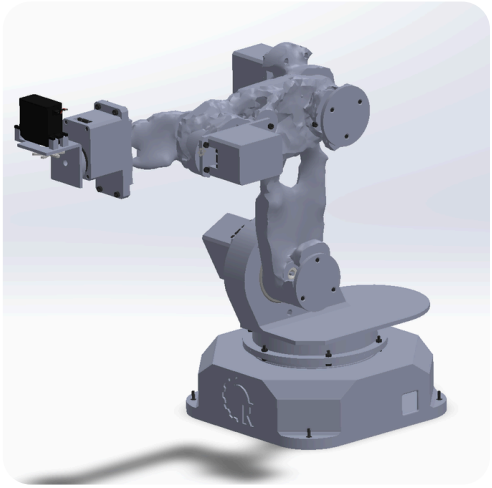
(Pre-Optimization)

(Optimized)

Final Design:



(Optimized)



(V1 Final)

During the design phase, I kept my engineering requirements in mind as I worked through several robotic-arm problems. These included non-concentric mounting holes, joint alignment, and components that would be difficult to manufacture. I originally hollowed out the links to reduce weight, but after starting the topology studies, I realized I was removing material before fully understanding where I needed it. I went back to a solid design and used topology optimization in ANSYS to identify the main load paths under the estimated maximum loading conditions. The resulting geometry helped me better understand how the loads traveled through each link and where material was needed.

Another design consideration was making the robotic arm easier to assemble and maintain. My original idea was to use 25 lb magnets to hold the joints in place and simplify mounting. However, I realized this could become a problem since I was only allowing two magnets per joint. To improve the mounting design, I added four 3.2 mm holes to each joint, so M3 screws could prevent unwanted rotation around the joint's center axis.

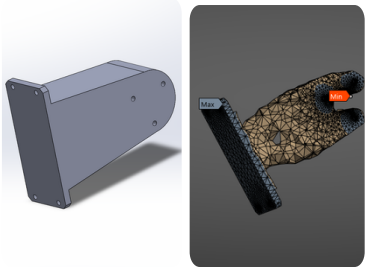
For the final V1 design, I used ANSYS topology optimization to reduce each link's mass while preserving key mounting features, including screw holes, nut pockets, and bearing inserts. I also adjusted the optimized geometry to make the components easier to manufacture and assemble in future versions.

System Analysis:

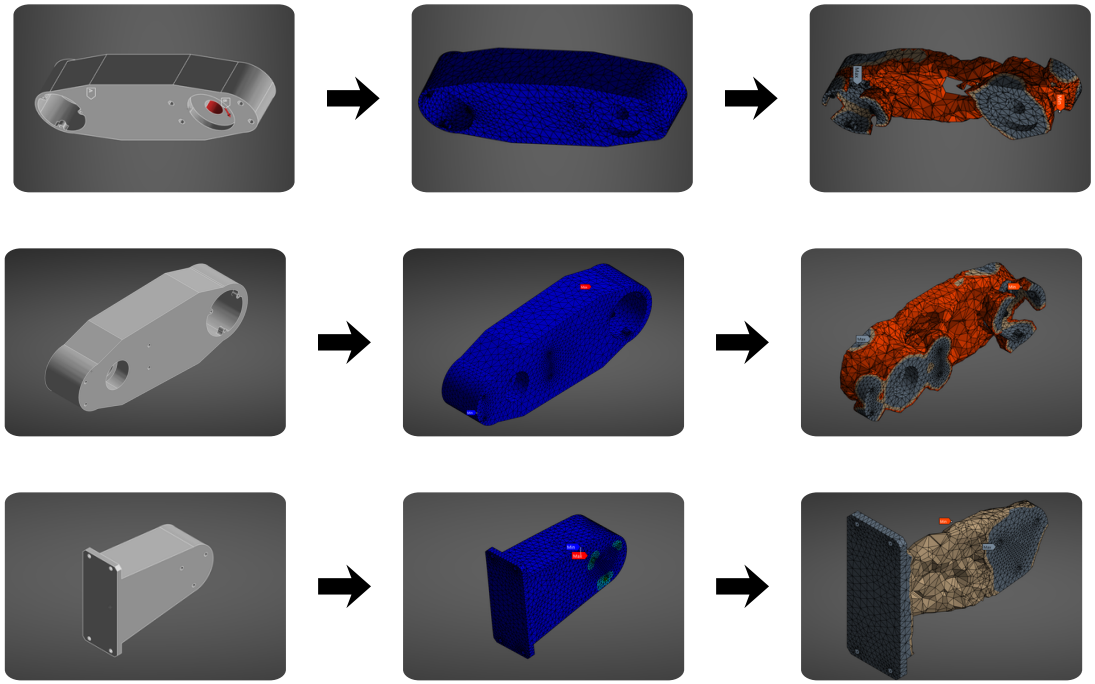
Design 1 Optimization:


(J1 to J2 Link)


(J2 to J3 Link)


(J4 to J5 Link)

Final V1 Design Optimization:


(Raw) (FEA) (Optimized)

As I continued developing each subassembly, I used ANSYS to perform structural analysis and topology optimization on the robotic arm's links. I created a custom PETG material profile using Elegoo's material properties and considered how FDM printing would affect the final parts' strength. Since 3D-printed components are anisotropic, I also considered the printing orientation so the main load paths would rely less on interlayer strength.
For each link, I used a target factor of safety of 2.0 when evaluating the estimated loading conditions. After the initial structural analysis, I used topology optimization to identify areas where I could remove material without significantly affecting the main load paths. I aimed to retain approximately 40% of the original mass while preserving important mounting features.

After each optimization study, I used the resulting geometry as a reference to refine the CAD models and improve their surfaces. I then checked the updated designs against the stress and deflection requirements to evaluate whether the optimized geometry could be used for the final V1 assembly.

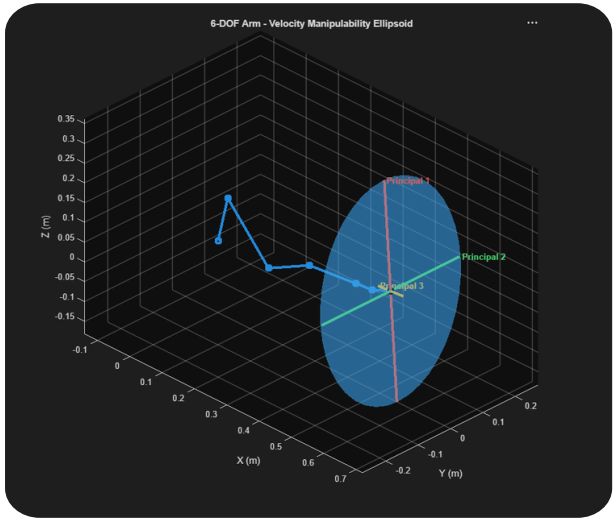
PETG Profile:

- ρ = 1260 kg/m^3
- E = 1.072 GPa
- v = 0.39 (PETG assumption)
- Tensile >= 51 MPa
- Flexural = 64 MPa
- Impact = 4.8 kJ/m^2

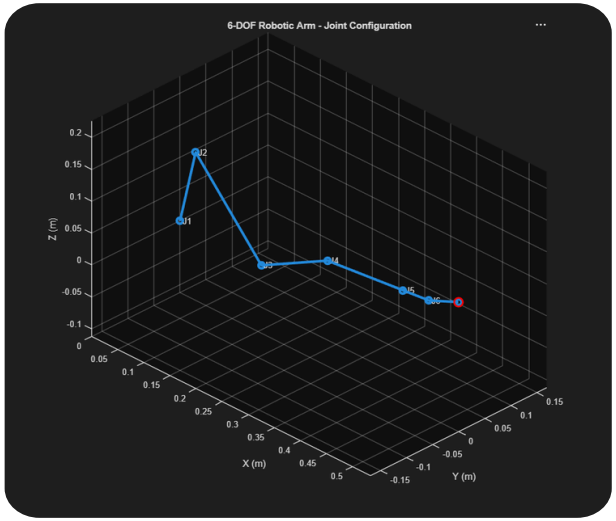
Iterations:

	Problem	Design Change	Rationale
Rev. A	J5 Servo did not fit properly inside the housing.	Reduced the main housing height by 4mm.	Allows the servo to sit flush with the mounting flange, improving fit and alignment.
Rev. B	Joint flanges exhibit excessive predicted deflection > 5mm.	Reduced the M3 insert hole diameter to 3.2mm	Increases the surrounding material thickness to improve flange stiffness and reduce defelction.
Rev. C	J4 servo does not contain a magnetic connector.	Added mounting hole for magnet and re-optimized J4 to J5 link.	Adding the magnetic connector allows for easy maintenance and assembly in the future.
Rev. D	Improper flange engagement at J4 connecting link J4 to J5.	Added longer mounting legs.	Adding longer legs stabalizes the joint while screwing in the magnet.
Rev. E	J2 flange has improper tolerance with bearing.	Switched 0.6mm tolerane to 0.2mm tolerance.	Shortening the toleance ensures proper engangement along the entire inner diameter of the bearing.

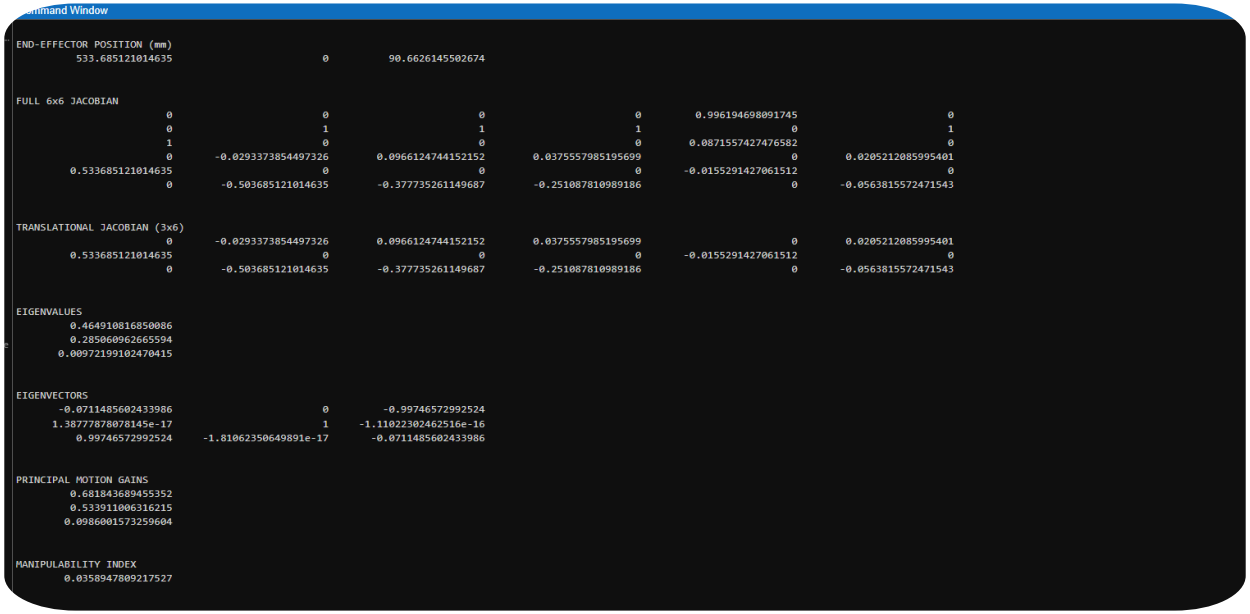
Calculations For All Six Joints: Joint Angles [0 45 -70 20 0 -15]



(Figure 1)



(Figure 2)



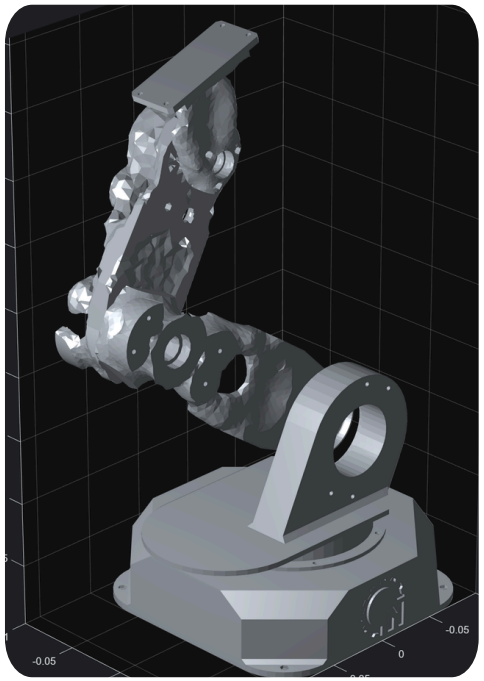
(Figure 3)

Performing kinematic calculations was an important part of understanding how my robotic arm moves. Using MATLAB, I calculated the Jacobian matrix to determine how changes in joint angles affect the end-effector's velocity and position. I then used eigenvalue and eigenvector analysis to identify the principal directions of movement and how effectively the arm could move in each direction.

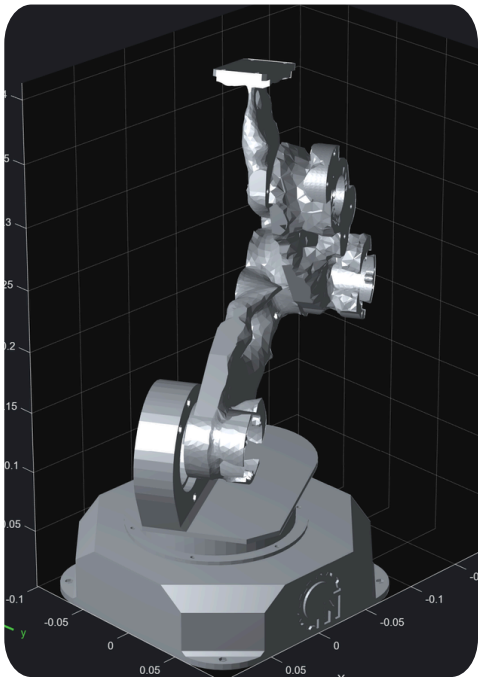
Using the joint angles [0 45 -70 20 0 -15], I calculated the Jacobian matrix and its eigenvalues and eigenvectors. Figure 1 shows the velocity manipulability ellipsoid, where the longer axes represent directions with better motion capability and the shorter axes represent more limited movement. Figure 2 shows the resulting joint configuration, while Figure 3 displays the calculated Jacobian, eigenvalues, eigenvectors, and principal motion gains.

Simulation:

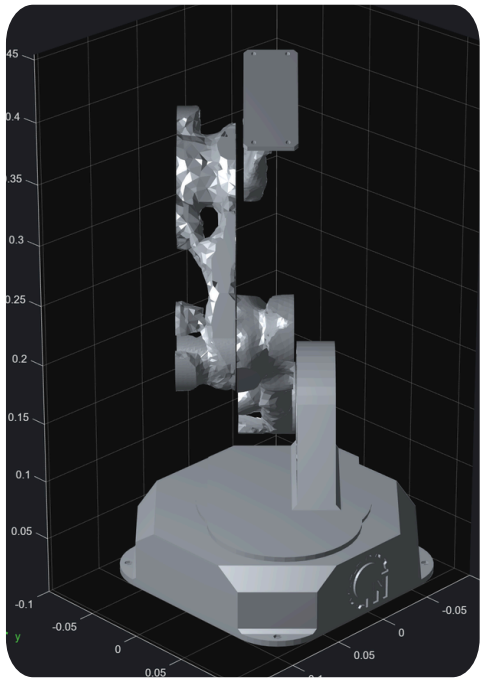
To simulate the robotic arm's movement, I used MATLAB's Robotics System Toolbox to develop a kinematic model using my optimized SOLIDWORKS components. I imported the STL files and manually configured the joint positions, rotation axes, and transformations. Once the links were aligned, I simulated the motion of joints J1 through J4 using different joint-angle configurations. I also implemented smooth motion interpolation to gradually move the arm between positions and plotted the joint angles throughout the simulation. This let me visualize the arm's range of motion, see how each joint affects overall movement, and evaluate the arm's kinematic behavior.



[90 55 -85 50] Degrees

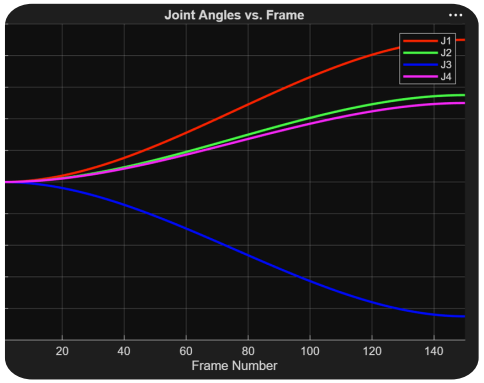


[-60 65 -105 55] Degrees

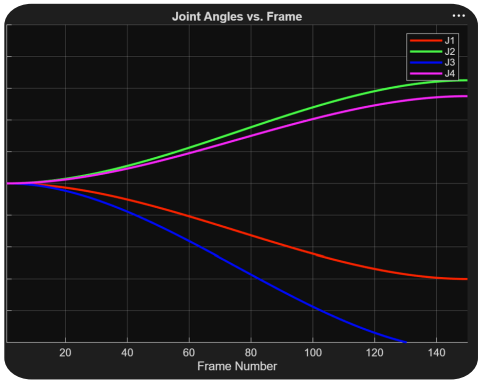


[135 50 -90 70] Degrees

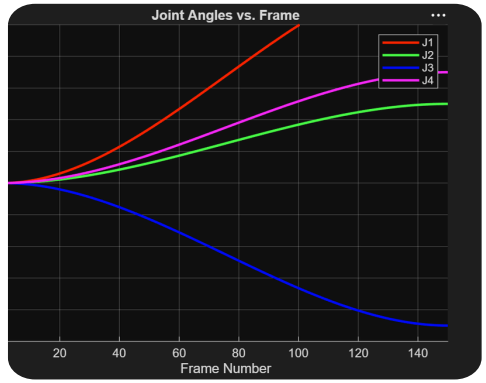
(MATLAB Simulated Joint Angles)



[90 55 -85 50]



[-60 65 -105 55]



[135 50 -90 70]

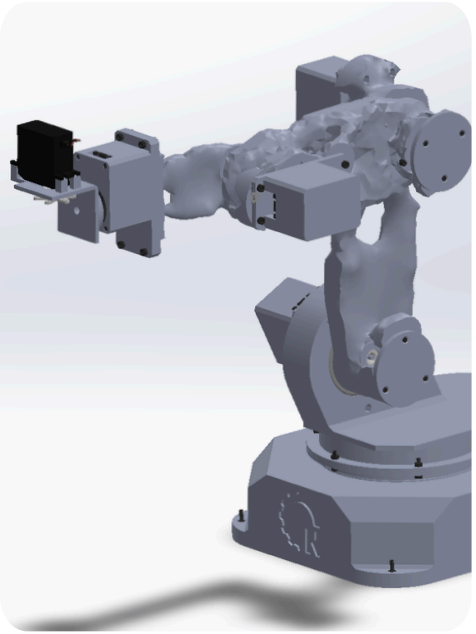
Results & Future Development:

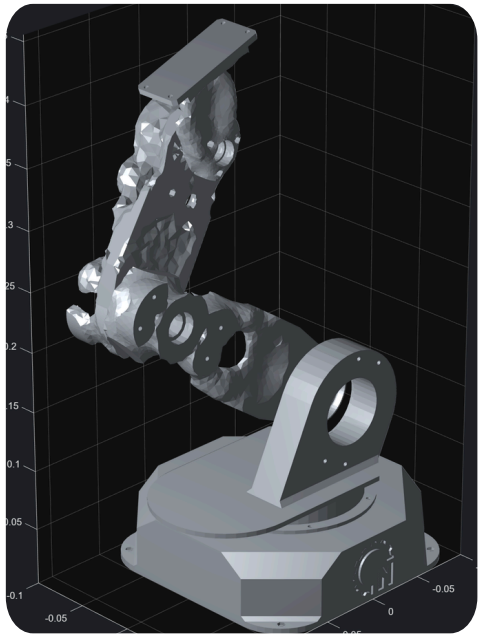
For Version 1, I focused on the mechanical design, structural optimization, and simulation of my 6-DOF robotic arm. Using SOLIDWORKS and ANSYS, I designed and optimized the mechanical components to improve their strength-to-weight ratio while maintaining important mounting features. I also tested several components' tolerances and revised the design to improve fit and overall assembly.

For the kinematic analysis, I developed a mathematical model of the entire 6-DOF robotic arm and calculated the Jacobian matrix to better understand the relationship between joint velocities and end-effector motion. I then used MATLAB's Robotics System Toolbox to simulate joint movements for J1 through J4. By implementing smooth motion interpolation, I simulated different joint configurations and visualized how the robotic arm moves.

For future development, Version 2 will focus on physically assembling the robotic arm, integrating the actuators, and testing the joint movements. Version 3 will focus on developing and integrating a mechanical gripper so the robotic arm can perform basic object manipulation.

Overall, this project helped me better understand mechanical design, structural analysis, and kinematic modeling. I also gained experience troubleshooting joint alignment, evaluating design tolerances, and improving the design throughout the process.

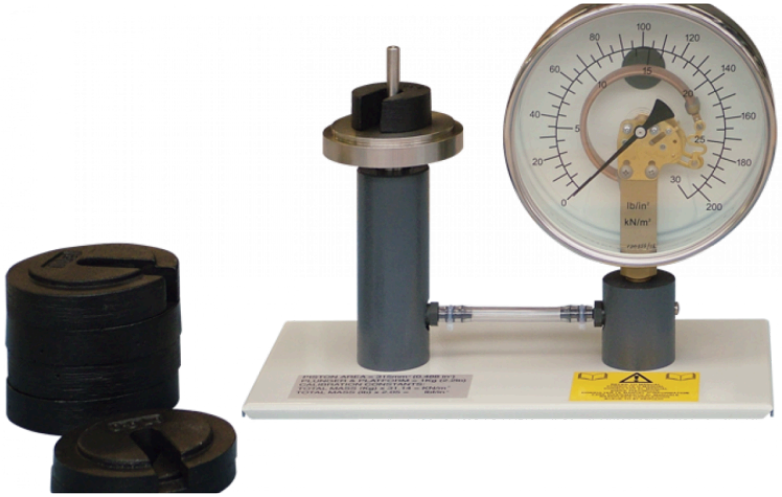


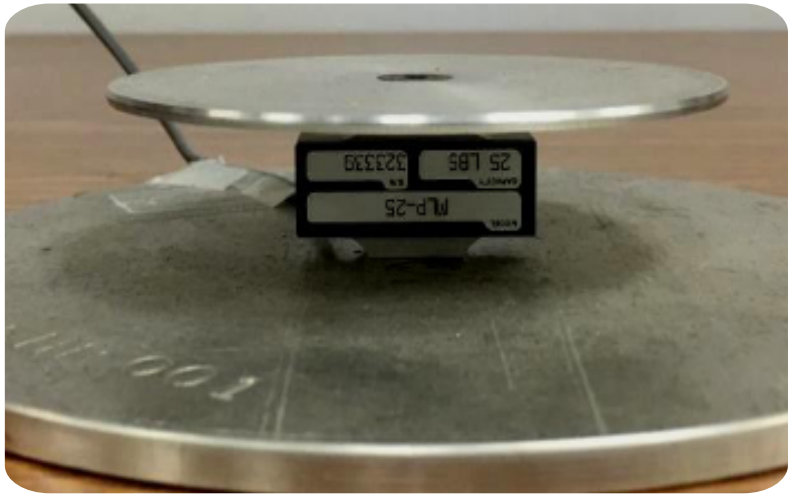


Engineering Requirements Satisfaction:

1. 6 DOF Design: ✓
2. Structural Analysis & Optimization: ✓
3. Component Tolerance Testing: ✓
4. Jacobian & Kinematic Calculations: ✓
5. J1-J4 Motion Simulation: ✓
6. Physical Assembly & Actuator Testing: **Planned V2**
7. Functional Gripper: **Planned V3**

Bourdon Gauge & Load Cell Calibration





Project Overview: This experiment aims to learn calibration methods and techniques using known input values. We took loading and unloading measurements at five input levels for each instrument and calculated the average outputs. We used the method of least squares to determine the correlation coefficients and standard errors, and used the regression line to find the calibration equations for the loading, unloading, and average data sets.

Tools: Bourdon Gauge • Load Cell • Excel

Raw Data:

Table 1: Bourdon Gauge

Input (PSI)	Loading (PSI)	Unloading (PSI)	Average (PSI)
200	670	680	675
400	1095	1095	1095
600	1500	1505	1502.5
1100	2520	2540	2530
1600	3520	3520	3520

Table 2: Load Cell

Input (kg)	Loading (kg)	Unloading (kg)	Average (kg)
1kg	-0.9789	0.0032	-0.48785
2kg	-1.9627	-0.9761	-1.4694
3kg	-2.9524	-1.9681	-2.46025
4kg	-3.9281	-2.9515	-3.4398
5kg	-4.9183	-3.9277	-4.423

Sample Calculations:

Table 3: Input vs. Loading

n	Input (x)	Loading (y)	XY	X^2	Y^2	$(Y-a_1 \times X-a_0)^2$
1	200	670	134000	40000	448900	145.008
2	400	1095	438000	160000	1199025	41.22677
3	600	1500	900000	360000	2250000	23.84897
4	1100	2520	2772000	1210000	6350400	72.93797
5	1600	3520	5632000	2560000	12390400	60.88361
Sum	3900	9305	9876000	4330000	22638725	343.9053

Correlation Coefficient:

$$R = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{\sqrt{\left(n\sum x_i^2 - (\sum x_i)^2\right)\left(n\sum y_i^2 - (\sum y_i)^2\right)}} = \frac{5(9876000)-(3900)(9305)}{\sqrt{(5(4330000)-(3900)^2)(5(22639725)-(9305)^2)}} = 0.9998737592$$

Calibration Equation:

$$y = a_0 + a_1 x$$

$$a_0 = \frac{\sum x_i \sum x_i y_i - \sum y_i \sum x_i^2}{(\sum x_i)^2 - n\sum x_i^2} = \frac{3900(9876000)-9305(4330000)}{\left(3900^2\right)-5(4330000)} = 275.5046584$$

$$a_1 = \frac{\sum x_i \sum y_i - n\sum x_i y_i}{(\sum x_i)^2 - n\sum x_i^2} = \frac{3900(9305)-5(9876000)}{(3900)^2-5(4330000)} = 2.032686335$$

$$y = 275.5046584 + 2.032686335x$$

Standard Error:

$$S_E = \left[\frac{\sum\left(y_{observed} - y_{predicted}\right)^2}{n-2}\right]^{\frac{1}{2}} = \sqrt{\frac{343.9053}{3}} = 10.706$$

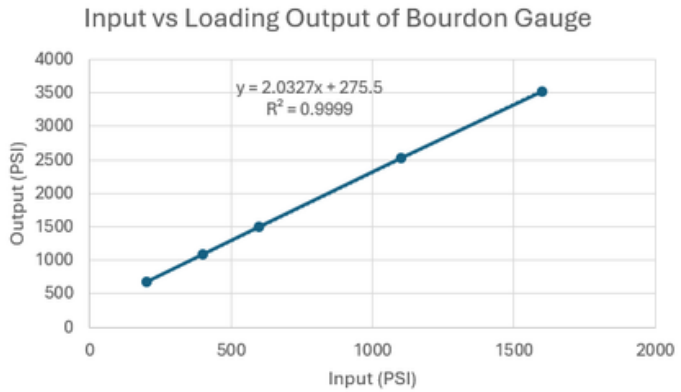
Coefficient of Correlation & Standard Error:

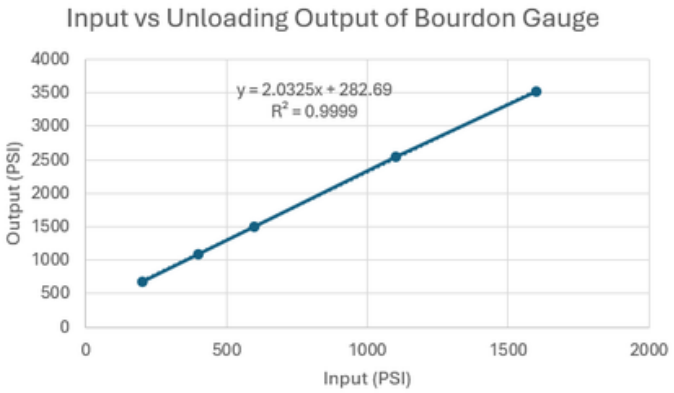
n	R	Calibration Equation	Error
BG Loading (PSI)	0.999873	y=275.504+2.032x	10.706
BG Unloading (PSI)	0.9999273	y=282.686+2.032x	16.056
BG Avg (PSI)	0.999954	y=279.095+2.032x	12.763
LC Loading (KG)	-0.9999978	y=0.0050-0.984x	0.003695
LC Unloading (KG)	-0.9999967	y=0.987-0.983x	0.004568
LC Avg (KG)	-0.99999866	y=0.496-0.9841x	0.0029335

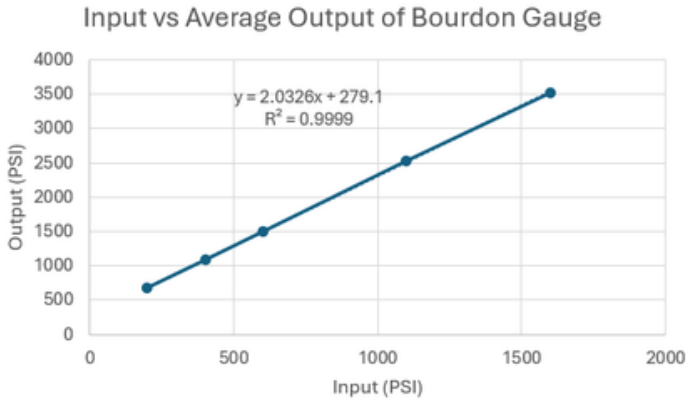
(Bourdon Gauge)

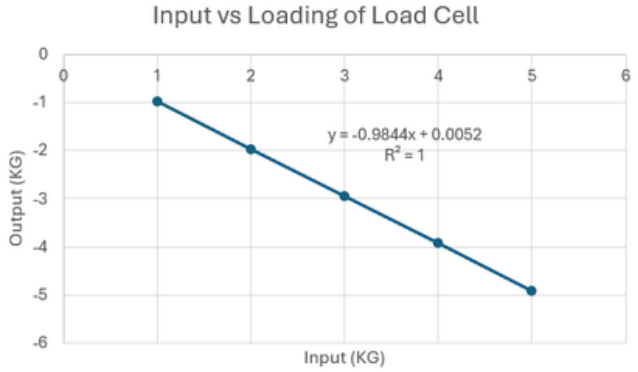
Calibration Curves:

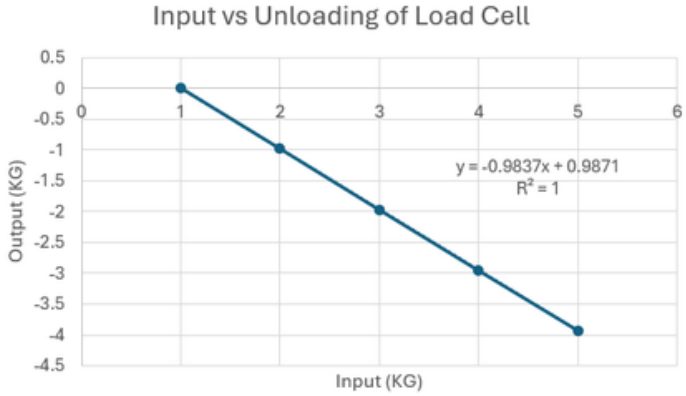
(Load Cell)

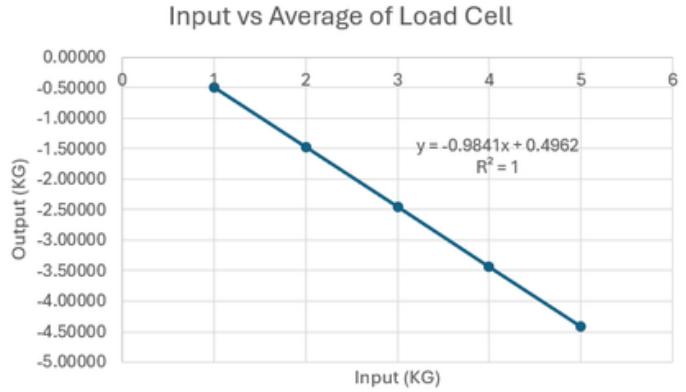












Conclusion:

This experiment used two instruments to measure weight: a Bourdon gauge and a load cell. The Bourdon gauge read pressure in PSI, and the reading rose each time a weight was added, reaching 3520 PSI with all the weights loaded. The gauge also had a large zero error. The fitted line reads about 279 PSI at zero input, and the readings on the way down did not retrace the way-up readings. For example, at a 200 PSI input, the gauge read 680 PSI on unloading and 670 PSI on loading. We applied the method of least squares to find the calibration line, correlation coefficient, and standard error. The averaged Bourdon data gave y = 279.1 + 2.0326x, with a correlation coefficient of 0.99995 and a standard error of 12.763 PSI, indicating a very close linear relationship.

The load cell measures weight with strain gauges. When a weight is placed on the platform, it bends, and the strain gauges measure the resulting compression and tension. With all five weights loaded, the load cell read -4.9183 kg. Its last unloading reading was 0.0032 kg, close to zero but not exactly zero. The same least squares method gave y = 0.4963 - 0.9841x for the averaged data, with a correlation coefficient of -0.999999 and a standard error of 0.0029 kg. The line is essentially perfectly linear, and its slope is close to 1 in magnitude.

Human error also affected the data, mostly on the Bourdon gauge. Adding the weights quickly gave a different pressure reading than adding them in small steps. The needle usually rested between two marks, so we had to estimate each reading by eye.